

Python Practical File

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Subject: Python Programming

Topic: Pattern Printing

Program: P5.1.py

```
# Program to print star pattern
for i in range (1, 6):
    for j in range(1, i + 1):
        print("*", end=" ")
    print()
```

Topic: Pattern Printing

Program: P5.2.py

```
#Program to print number pattern
for i in range(1, 6):
    for j in range(1, i + 1):
        print(j, end= " ")
    print()
```

Topic: Pattern Printing

Program: P5.3.py

```
#Program to print pyramid pattern
n = 5
for i in range(1, n + 1):
    print(" " * (n - i), end=" ")
    print("* " * i)
```

Topic: Pattern Printing

Program: P5.4.py

```
# Program to print alphabet triangle pattern
for i in range(65, 70):
    for j in range(65, i + 1):
        print(chr(j), end=" ")
    print()
```

Topic: Pattern Printing

Program: P5.5.py

```
# Program to print alphabet triangle pattern
n = 5
for i in range(n):
    num = 1
    for j in range(n - 1):
        print(" ", end="")
    for j in range(i + 1):
        print(num, end=" ")
        num = num * (i - j) // (j + 1)
```

```
print()
```

Topic: Python Conditional Statements

Program: P3.1.py

```
#Program using simple if statement

a = int(input("Enter The Value Of A"))
b = 50

if (a > b):
    print("A Is Greater Than B")

if (a == b):
    print("A Is Equal To B")

if(a < b):
    print("A Is Smaller Than B")
```

Topic: Python Conditional Statements

Program: P3.2.py

```
# Program using ifelse statement

a = 45
b = 35

if(a > b):
    print("A Is Greater Than B")
else:
    print("A Is Smaller Than B")
```

Topic: Python Conditional Statements

Program: P3.3.py

```
# Program using nested ifelse

a = 45
b = 45

if(a > b):
    print("A Is Greater Than B")
else:
    if(a < b):
        print("A Is Smaller Than B")
    else:
        print("A Is Equal To B")
```

Topic: Python Conditional Statements

Program: P3.4.py

```
# Program using elif ladder

a = int(input("Enter Value 1"))
b = int(input("Enter Value 2"))

if(a > b):
    print("A Is Greater Than B")
elif(a < b):
    print("A Is Smaller Than B")
else:
    print("A Is Equal To B")
```

Topic: Python Data Types & Variables

Program: P1.1.py

```
print("Hello World!")
```

Topic: Python Data Types & Variables

Program: P1.2.py

```
# Program to declare and print different types of variables
age = 20
height = 5.9
grade = 'A'
salary = 25000.75
name = 'Prachi Sharma'
is_student = True

print("Integer: ", age)
print("Float (height): ", float(height))
print("Grade: ", str(grade))
print("Salary: ", float(salary))
print("Name: ", str(name))
print("Boolean ", bool(is_student))
```

Topic: Python Data Types & Variables

Program: P1.3.py

```
#3. Program to take user input and display its data type
# Note: In Python, input() always takes data as string (str) by default

user_input = input("Enter Something: ")
try:
    val = int(user_input)
    print("Value Is Integer")
except:
    try:
        val = float(user_input)
```

```

        print("Value Is Float")
    except:
        if user_input.lower() in ["true", "false"]:
            print("Data Type: Boolean")
        else:
            print("Data Type: String")

```

Topic: Python Data Types & Variables

Program: P1.4.py

```

val1 = 45
val2 = 75

temp = val1
val1 = val2
val2 = temp

print("After Swapping: ")
print("a= ", val1)
print("b= ", val2)

```

Topic: Python Data Types & Variables

Program: P1.5.py

```

# Program to assign multiple variables in a single line.

a, b, c = 10, 20, 30
print(a, b, c)

```

Topic: Python Lists

Program: P8.1.py

```

# Program to create and access list elements
my_list = [10, 20, 30, 40, 50]

print("List: ", my_list)
print("First Element " ,my_list[0])
print("Second Element: ", my_list[1])
print("Last Element: ", my_list[-1])

for item in my_list:
    print(item)

```

Topic: Python Lists

Program: P8.2.py

```

# Program demonstrating list slicing
my_list = [10, 20, 30, 40, 50, 60]

```

```
# List slicing
print("Original list:", my_list)
print("First 3 elements:", my_list[:3])
print("Elements from index 2 to 4:", my_list[2:5])
print("Last 3 elements:", my_list[-3:])
print("Every second element:", my_list[::2])
```

Topic: Python Lists

Program: P8.3.py

```
# Program using built-in list methods
list = [10, 20, 30]
list.append(40)
list.insert(1, 15)
list.remove(20)
list.pop()
list.sort()

print("Final List: ", list)
```

Topic: Python Lists

Program: P8.4.py

```
list = [10, 20, 30, 40]
total = sum(list)
print(total)
```

Topic: Python Lists

Program: P8.5.py

```
# Program to find largest and smallest element in a list
list = [10, 40, 80, 30, 20]
print("Maximum: ", max(list))
print("Minimum: ", min(list))
```

Topic: Python Lists

Program: P8.6.py

```
# Program to remove duplicate elements from a list
lst = [10, 20, 30, 20, 10, 40]
unique = list(set(lst))
unique.sort()
print(unique)
```

Topic: Python Lists

Program: P8.7.py

```
# Program to sort a list
```

```
list = [10, 40, 30 , 50, 20]
list.sort()
print(list)
list.sort(reverse=True)
print(list)
```

Topic: Python Looping Statements

Program: P4.1.py

```
# Program using for loop

for i in range(1, 6):
    print(i)
# 1
# 2
# 3
# 4
# 5

for i in range(1, 6, 2):
    print(i)

# 1
# 3
# 5
sum = 0
for i in range(1, 6):
    sum += i
print(sum)
```

Topic: Python Looping Statements

Program: P4.2.py

```
#Program using while loop

i = 1
total = 0
while(i <= 5):
    total += i
    print(total)
    i += 1
```

Topic: Python Looping Statements

Program: P4.3.py

```
#Program using nested for loop
for i in range(1, 6):
    for j in range(1, 6):
        print("*", end= " ")
    print()
```

Topic: Python Looping Statements

Program: P4.4.py

```
# Program using break statement
for i in range(1, 10):
    if i == 5:
        break
    print(i)
```

Topic: Python Looping Statements

Program: P4.5.py

```
# Program using continue statement
# Skip number 5
for i in range (1, 10):
    if i == 5:
        continue
    print(i)
```

Topic: Python Operators

Program: P2.1.py

```
a = 25
b = 344
c = a + b
print(c)
```

Topic: Python Operators

Program: P2.2.py

```
a = input("Enter Value 1 ")
b = input("Enter Value 2 ")

print("a == b", a == b)
print("a != b", a != b)
print("a >= b", a >= b)
print("a <= b", a <= b)
print("a > b", a > b)
print("a < b", a < b)
```

Topic: Python Operators

Program: P2.3.py

```
a = int(input("Enter Value 1 "))
b = int(input("Enter Value 2 "))

print("a > 0 and b > 0: ", a > 0 and b > 0)
print("a > 0 or b > 0", a > 0 or b > 0)
```

```
print("not(a > b)", not(a > b))
```

Topic: Python Operators

Program: P2.4.py

```
# Initial value
x = 10

print("Initial value:", x)

# Using assignment operators
x += 5    # x = x + 5
print("After += :", x)

x -= 3    # x = x - 3
print("After -= :", x)

x *= 2    # x = x * 2
print("After *= :", x)

x /= 4    # x = x / 4
print("After /= :", x)

x %= 3    # x = x % 3
print("After %= :", x)
```

Topic: Python Strings

Program: P6.1.py

```
# Program to create and index a string
s = "Python"

print("String", s)
print("First Character: ", s[0])
print("Last Character: ", s[-1])
```

Topic: Python Strings

Program: P6.2.py

```
# Program demonstrating string slicing

s = "Pythonnn"

print("Original", s)
print("Slice(0 : 6): ", s[0:6])
print("Slice(3:): ", s[3:])
print("Slice(:5): ", s[:5])
```

Topic: Python Strings

Program: P6.3.py

```
#Program using built-in string methods
s = "Hello World"

print(s.upper())
print(s.lower())
print(s.title())
print(s.replace("world", "hello"))
```

Topic: Python Strings

Program: P6.4.py

```
#Program to check palindrome string

s = input("Enter String ")
if s == s[::-1]:
    print("Palindrome")
else:
    print("Not Palindrome")
```

Topic: Python Strings

Program: P6.5.py

```
# Program to find length of a string
s = input("Enter String ")
print(len(s))
```

Topic: Python Strings

Program: P6.6.py

```
# Program to reverse a string

s = input("Enter String: ")
print("Reversed String: ", s[::-1])
```

Topic: Python Strings

Program: P6.7.py

```
# Program to count vowels in a string

s = input("Enter String: ")
count = 0
for ch in s.lower():
    if ch in 'aeiou':
        count += 1

print("Number Of Vowels: ", count)
```

Topic: Python Strings

Program: P7.1.py

```
# Program to Count Occurrence of a Character
s = input("Enter A String ")
ch = input("Enter Char To Count ")

count = s.count(ch)
print("Occurences: ", count)
```

Topic: Python Strings

Program: P7.2.py

```
# Program to Find Position of a Character
s = input("Enter A String ")
ch = input("Enter Char To Find: ")

pos = s.find(ch)
print("Posotion Of Character Is: ", pos)
```

Topic: Python Strings

Program: P7.3.py

```
# Program to Check Whether String Starts with a Word
s = input("Enter A String ")
word = input("Enter Starting Word")
eword = input("Enter Ending Word: ")
print(s.startswith(word))
print(s.endswith(eword))
```

Topic: Python Strings

Program: P7.4.py

```
# Program to Split a String

s = input("Enter A String ")
words = s.split()
print("SplitList: ", words)
```

Topic: Python Strings

Program: P7.5.py

```
# Program to Join Strings
words = ["Python", "is", "easy"]
result = " ".join(words)
print(result)
```

Topic: Python Tuples

Program: P9.1.py

```
# Program to create and access tuple
t = (10, 20, 30, 40)
print("Tuple", t)
print("First Element: ", t[0])
print("Last Element: ", t[-1]);
```

Topic: Python Tuples

Program: P9.2.py

```
# Program demonstrating tuple slicing
t = (10, 20, 30, 40, 50, 60)

print("Original Tuple: ", t)
print("First Three Elements: ", t[:3])
print("Middle Elements: ", t[1:4])
print("Last 2 Elements", t[-2:])
```

Topic: Python Tuples

Program: P9.3.py

```
# Program to convert tuple to list

t = (10, 20, 30, 40)

lst = list(t)
print("Tuple: ", t)
print("List: ", lst)
```

Topic: Python Tuples

Program: P9.4.py

```
# Program to find length of a tuple
t = (10, 20, 30, 40, 50)
length = len(t)
print(length)
```

Topic: Python Tuples

Program: P9.5.py

```
# Program to concatenate two tuples

t1 = (10, 20, 30)
t2 = (40, 50, 60)

t3 = t1 + t2
```

```
print(t3)
```

Topic: Sets And Dictionary

Program: P10.1.py

```
# Program to perform set operations
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}

print("Union: ", A | B)
print("Intersection: ", A & B)
print("Difference (A - B)", A - B)
print("Symmetric Difference: ", A ^ B)
```

Topic: Sets And Dictionary

Program: P10.2.py

```
# Program using set methods
s = {1, 2, 3, 4}
s.add(5)
s.remove(4)
s.discard(5)
s.pop()

print("Final Set: ", s)
```

Topic: Sets And Dictionary

Program: P10.3.py

```
# Program to create and print a dictionary

student = {
    "name": "Aaditya",
    "age": 20,
    "course": "BCA"
}

print("Dictionary: ", student)
```

Topic: Sets And Dictionary

Program: P10.4.py

```
student = {
    "name": "Aaditya",
    "age": 28
}
student["course"] = "BCA"
print(student)
```

Topic: Sets And Dictionary

Program: P10.5.py

Program to update and modify dictionary values

```
student = {  
    "name": "Prachi Sharma",  
    "age": 20  
}  
student["age"] = 21  
student["course"] = "B.Sc Nursing"  
print(student)
```

Topic: Sets And Dictionary

Program: P10.6.py

Program to delete elements from a dictionary

```
student = {  
    "name": "Prachi Sharma",  
    "age": 19,  
    "course": "B.Sc Nursing"  
}  
student.pop("age")  
del student["course"]  
# student.clear() // Remove All Values And Keys  
print(student)
```